

Traffic Signal Violation Detection at Simpang Wirosaban Using YOLOv11, HSV Thresholding, and SORT

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Abstract—This is a placeholder abstract. Replace this text with a concise summary of the research background, method, main findings, and conclusion.

I. INTRODUCTION

A. Background

Traffic signal violations, particularly red light running, constitute one of the most prevalent and dangerous forms of traffic misconduct at urban intersections. [1] In Indonesia, the problem has reached a critical scale. Data from the National Criminal Information Center of the Indonesian National Police recorded 148,798 traffic accidents throughout 2023, of which 94.63% were attributed to human error [2]. The Director of Traffic Accident Investigation of the Indonesian National Police further stated that every accident is preceded by a violation, underscoring the direct causal relationship between traffic infractions and collision events [2]. Simpang Wirosaban, a signalized intersection in Yogyakarta, represents a typical urban intersection in Indonesia where such violations occur at high frequency, particularly during nighttime hours when traffic enforcement is minimal. The absence of automated monitoring at such locations necessitates a technology-driven approach to continuous, objective violation detection.

Conventional methods of traffic enforcement rely on human officers stationed at intersections or on radar-based sensors embedded in road infrastructure. These approaches are costly to deploy and maintain, limited in spatial coverage, and susceptible to human fatigue and inconsistency [3]. The proliferation of closed-circuit television (CCTV) infrastructure at intersections across Indonesian cities presents an opportunity to repurpose existing hardware for automated enforcement

without additional installation costs. Computer vision techniques applied to CCTV footage have demonstrated considerable promise in this domain. Ong et al. [3] proposed a YOLO-based system targeting red light crossing and double-line overtaking violations from CCTV footage, establishing a foundational framework for vision-based violation detection. Similarly, Tomas et al. [4] demonstrated that YOLOv5 combined with Faster R-CNN achieved high accuracy in detecting red light and wrong-way violations at intersections, reinforcing the viability of deep learning pipelines for this task.

Vehicle detection in video-based systems has been significantly advanced by the YOLO family of single-stage object detectors. Since its introduction, YOLO has undergone continuous architectural refinement, with each iteration improving detection accuracy and inference speed. YOLO11, released by Ultralytics in September 2024 at YOLO Vision 2024, introduced C3K2 blocks, an enhanced Spatial Pyramid Pooling Fast (SPPF) module, and a C2PSA partial spatial attention mechanism [5]. The YOLO11m variant achieves higher mean average precision on the COCO benchmark with 22% fewer parameters than its YOLOv8m predecessor, making it well-suited for real-time deployment on constrained hardware [5]. For multi-object tracking across video frames, this study employs the Simple Online and Realtime Tracking (SORT) algorithm proposed by Bewley et al. [6], which combines the Kalman filter for state estimation with the Hungarian algorithm for frame-to-frame data association. Despite its computational simplicity, SORT achieves tracking performance comparable to significantly more complex methods while operating at over 260 Hz, satisfying the latency requirements of real-time traffic monitoring.

A persistent challenge in nighttime traffic surveillance is the degradation of image quality caused by uneven illumination, vehicle headlight glare, and sensor noise, all of which reduce object detection confidence. Lashkov et al. [7] demonstrated that Contrast Limited Adaptive Histogram Equalization (CLAHE) applied as a preprocessing step substantially improves vehicle detection performance in nighttime conditions. Unlike standard histogram equalization, CLAHE operates on localized image tiles and enforces a contrast limit to prevent noise amplification, making it particularly effective for scenes with spatially non-uniform lighting such as signalized intersections at night [8], [9]. In this study, CLAHE is applied exclusively to the luminance channel of the LAB color space prior to YOLO11 inference, preserving the chrominance information required by the HSV-based traffic light detector. This decoupled enhancement strategy ensures that preprocessing improvements to vehicle detection do not introduce artifacts into the color-sensitive components of the pipeline.

B. Objective

The primary objectives of this study include:

- To develop a real-time traffic violation detection system capable of identifying vehicles that cross the stop line during a red light signal using CCTV footage.
- To implement vehicle detection and multi-object tracking using YOLOv11 and SORT algorithm to accurately localize and track vehicles within a defined detection zone.
- To design a rule-based violation logic that determines infractions based on vehicle trajectory relative to the stop line and the current traffic light state.
- To evaluate the effectiveness of image enhancement techniques, including CLAHE, Gamma Correction.

C. Scope

The scope of the study includes:

- The study only can be implemented at the Simpang Wirosaban, Yogyakarta, using footage from a fixed CCTV camera.
- The system is optimized for nighttime conditions, where traffic light colors are more distinguishable.
- Detection performance may decrease under rainy weather, poor visibility, or varying lighting conditions.

II. LITERATURE REVIEW

A. Related Work

- 1) Traffic Signal Violation Detection at Simpang Gedongan Sleman Using YOLOv8, HSV-Based Segmentation, and DEEPSORT

This study addresses the inefficiency and human error associated with manual traffic enforcement. The research proposes a real-time automated surveillance system using YOLOv8 for vehicle identification and DeepSORT for trajectory tracking. A key contribution is the integration of the HSV (Hue, Saturation, Value) color model to detect traffic light states with high precision

under varying environmental lighting conditions. Result: The system demonstrates high robustness in distinguishing between "Red" and "Green" signals regardless of daylight or nighttime conditions. By utilizing a polygon-based Region of Interest (ROI) and a virtual stop line, the system effectively flags violators and generates visual outputs like bounding boxes and alert indicators.

- 2) Advanced Deep Learning for Traffic Surveillance

This research focuses on the application of deep learning architectures to maintain high detection accuracy in complex urban environments. The study emphasizes that while object detection identifies vehicles, the integration of advanced tracking algorithms is critical for consistent monitoring within a predefined Region of Interest (ROI). Result: The implementation of these models significantly improves the accuracy of identifying crossing violations by maintaining unique vehicle identities (IDs) even during occlusion. The findings suggest that digitizing the enforcement process through these combined methodologies enhances road safety and reduces the operational workload on traffic authorities.

B. HSV (Hue, Saturation, Value) Color Model

In the field of computer vision, selecting an appropriate color space is critical for the accuracy of object detection under varying environmental conditions. The HSV (Hue, Saturation, Value) color model is a cylindrical representation of colors that more closely aligns with human visual perception compared to the standard RGB model. While RGB defines colors as a combination of primary light sources, it is highly sensitive to changes in brightness, shadows, and external lighting. The HSV model decouples color information from intensity, using three distinct components:

- Hue (H): Represents the dominant wavelength or the actual "color" (e.g., Red, Yellow, Green) expressed in degrees from 0° to 360° .
- Saturation (S): Indicates the vibrancy or purity of the color, ranging from 0 to 100%.
- Value (V): Represents the brightness or luminance of the color.

C. You Only Look Once (YOLOv11)

D. Deep Simple Online and Realtime Tracking (SORT)

III. METHODOLOGY

A. Proposed Approach

This study proposes a computer vision-based approach for detecting traffic signal violations using CCTV at Wirosaban footage from a static camera. The system is designed as a multi-stage pipeline consisting of preprocessing, vehicle detection, object tracking, traffic light recognition, and violation analysis. To improve efficiency and reduce noise, a polygon-based Region of Interest (ROI) is defined to limit the analysis to the roadway area, excluding irrelevant regions such as sidewalks and surrounding infrastructure.

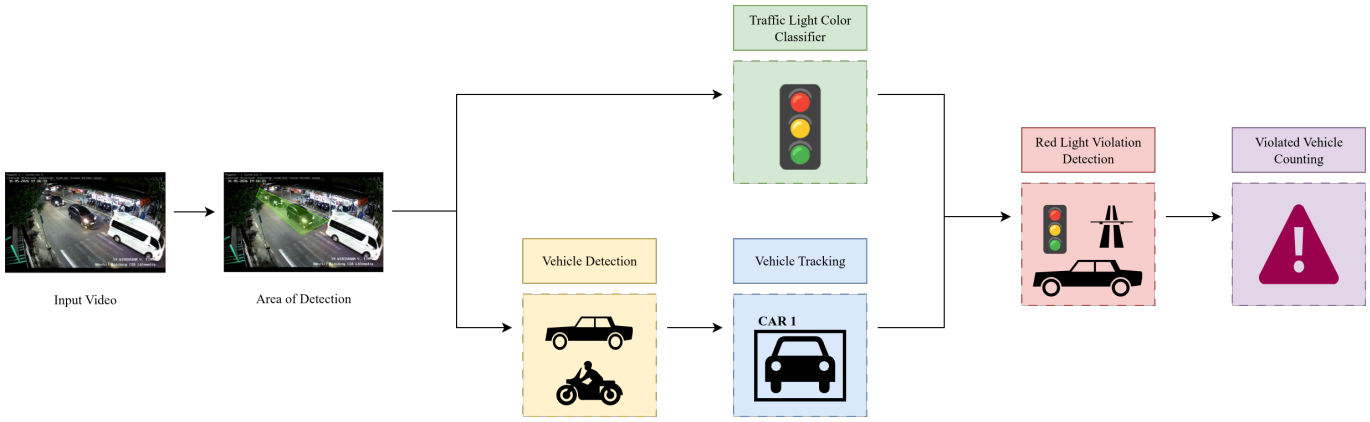


Fig. 1: Diagram flow of traffic signal violation detection at Simpang Wirosaban.

Within this ROI, vehicle detection is performed using YOLOv8, focusing only on vehicles located inside the polygonal area. The detected vehicles are then processed using an object tracking method to assign unique identities and maintain their trajectories across consecutive frames. This tracking process enables consistent monitoring of each vehicle's movement over time, which is essential for determining whether a vehicle has crossed a predefined boundary.

In parallel, the system identifies the traffic light state using a HSV color-based approach, which is used as a condition in the violation detection process. A virtual stop line is defined within the ROI to represent the legal stopping boundary. A violation is detected when a tracked vehicle crosses this line while the traffic light is in a red state. The system then generates an output in the form of visual annotations and alert messages, providing a structured and automated method for traffic violation detection.

B. Dataset

Write the dataset description here.

C. Preprocessing

Write the preprocessing steps here.

D. Detection and Tracking

Write the detection and tracking method here.

E. Violation Logic

Write the violation decision logic here.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Write the experimental setup here.

B. Results

Write the main results here.

V. CONCLUSION

A. Conclusion

Write the final conclusion here.

B. Future Work

Write possible future improvements here.

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